MODULE 3 OVERVIEW

This self-study module is designed to equip Operations Field Technicians with the knowledge and familiarization of the properties associated with LNG, how to be prepared to recognize process hazards, how LNG can produce cryogenic burns, how to handle cryogenic liquids, and how to wear the proper personal protective equipment (PPE).

The possibility of large spills of liquefied natural gas (LNG) from large, insulated storage tanks or ocean-going LNG tank ships has received increased attention as the demand for natural gas has increased. This attention has resulted because:

- LNG has hazardous properties,
- LNG is stored at extremely low temperatures in large, insulated storage tanks, and
- LNG is transported in large, highly visible ships.

LNG spills are considered serious events because the spill surface provides a large heat source that will rapidly vaporize the LNG. The resultant flammable vapor cloud can travel downwind before diluting to safe concentrations. If the flammable vapor is ignited, a vapor cloud fire followed by a pool fire can occur.

Learning objectives are provided to guide you to the major points in the material and to help you gauge your understanding of the content and your progress. To obtain maximum benefit, each person completing this self-study module should study the material covered and answer the review questions when finished. Each question is designed to address the listed associated learning objectives.

Learning Objectives for this Self-Study Module

1. State the chemical compounds of hydrocarbons.
2. Describe the physical properties of hydrocarbons.
3. Name the three states of matter.
4. Describe the properties of gases.
5. Describe the key definitions of the characteristics of LNG.
6. Discuss the fire hazards of LNG.
7. Describe the types of LNG hazards.

Module Layout

This self-study module provides the basis for training in your specific job responsibilities as they relate to LNG. To safely complete all assigned tasks, you must have an in-depth knowledge of how the process works and operates.

Each lesson in this module builds on the foundation established in the previous lesson. These lessons include a general description including the gas flow path, major equipment, and system operation and control.

Major equipment components are listed in this document with their proper name and Tag Number, i.e., Feed Gas Heater (5E101). Tag numbers are used to standardize identification of those components in drawings, process flow diagrams, P&IDs (Piping and Instrumentation Diagrams) (Piping and Instrumentation Diagrams) and on the actual equipment in the field.

Notes, Cautions, Warnings, and Environmental information may be listed throughout each lesson. This is critical information related to operations, personnel safety, and environmental protection. Read each as it is presented. Notes, Cautions, Warnings, and Environmental Information are described as shown in the legend below.

LEGEND

NOTE

A **NOTE** is used to provide additional information that may be helpful.

CAUTION!

A **CAUTION** is used when the possibility of equipment damage or a serious process upset may occur.

WARNING!

A **WARNING** is used when the possibility of injury or death exists.

Environmental Information

Environmental Information is used to provide specific information or references to Environmental Policy and/or Procedures.

At the conclusion of this self-study module, you will be given a test that encompasses the module information. You will be expected to achieve a minimum passing score.

Glossary of Terms

The glossary provides definitions of major terms used in this module. Familiarity with these terms will assist you in performing your job effectively.

TERM	DEFINITION
CFM	Cubic Feet per Minute
Cryogenic	In Physics, *Cryogenics* is the study of the production and behavior of materials at extremely low temperatures (below −292.00°F).
Emergency	An event that disrupts normal operating conditions and can pose a threat to human life, health, property, or the environment, if not immediately controlled.
Hydrocarbons	Substance made of hydrogen and carbon (among other elements). Common hydrocarbons found in Natural Gas (AKA "Fuel Gas" @ Cove Point LNG): • Methane • Ethane • Propane • Butane • Pentane
LNG	Liquefied Natural Gas: Natural gas composed primarily of methane that is cooled to a temperature of about −260°F at atmospheric pressure where it condenses to a liquid.
LPG	Liquefied Petroleum Gas (LPG), called simply "propane" or "butane," are flammable mixtures of hydrocarbon gases stored in pressurized containers.
MSDS/SDS	Material Safety Data Sheet / Safety Data Sheet
psi	Pounds per Square Inch
psig	Pounds per Square Inch (Gauge)

INTRODUCTION

This self-study module is an introduction to the characteristics of LNG. Understanding the overall function and characteristics of LNG provides you with the information required to work with LNG safely and economically.

In this module, we review the following main topics:

- Hydrocarbons
- States of Matter
- Properties of Gases
- Characteristics of LNG
- Physical Properties of Hydrocarbons
- Hazards of LNG
- Understanding LNG Fire Hazards
- Types of LNG Hazards

SAFETY NOTE

All personnel shall comply with the requirements of the Cove Point LNG *Accident Prevention Manual*. Specific hazards associated with this discussion are:

- Asphyxiation
- Fire
- Cryogenic Liquid (-238°F)

HYDROCARBONS

What are "hydrocarbons"? Any compound made from carbon and hydrogen is called a ***hydrocarbon***.

The carbon and hydrogen atoms are joined by chemical bonds to form hydrocarbon molecules.

Physical Properties of Hydrocarbons

The properties of each type of hydrocarbon depend on the number of carbon and hydrogen atoms in the hydrocarbon molecules, and how they are joined.

Hydrocarbon is the name given to the large family of chemical compounds made up entirely of hydrogen and carbon. Crude oil is a mixture composed almost entirely of hydrocarbons, with a few other compounds like hydrogen sulfide, H_2S, present in some cases.

Learning Objective 1

State the chemical compounds of hydrocarbons.

There are millions of different ways in which carbon and hydrogen atoms can be connected to form hydrocarbon molecules. To make it easier to classify them, these materials are grouped into several large families, e.g., the "paraffin" family."

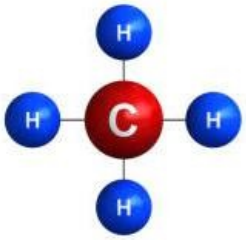
The smallest hydrocarbon molecule is **Methane**, which is the major constituent of natural gas. It has one carbon atom connected to four hydrogen atoms. Its chemical formula is represented as CH_4 and its molecular structure is shown below in **Figure 1**.

Learning Objective 2

Describe the physical properties of hydrocarbons.

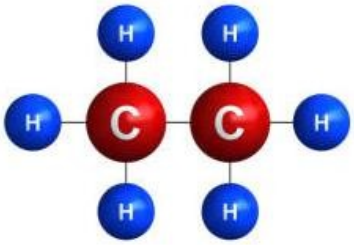
- In any hydrocarbon, the carbon atoms behave as though they each have four arms and the hydrogen atoms behave as though they each have one arm.
- Each arm of the carbon atom must always be occupied, i.e., it must be holding something, either a hydrogen atom, another carbon atom, or an atom of some other element.

- **Methane** The simplest member of the paraffin family.



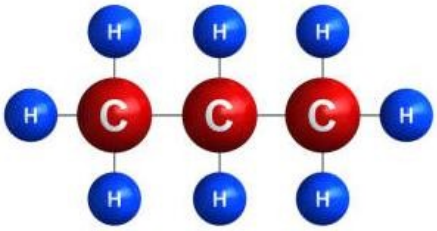
Methane CH_4

- **Ethane** By adding one carbon atom and two hydrogen atoms, methane is converted to **ethane**.



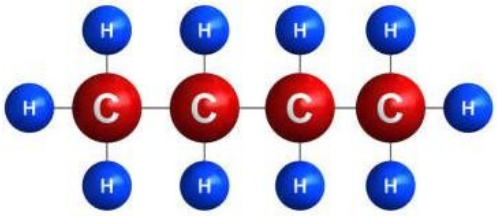
Ethane C_2H_6

- **Propane** If one more carbon atom and two more hydrogen atoms are added, the result is **propane**, the third member of the paraffin family.



Propane C_3H_8

- **Butane** The **butane** molecule is composed of four carbon atoms.



Butane C_4H_{10}

Figure 1: Physical Properties of Hydrocarbons

Boiling Point

The normal ***boiling point*** is "the temperature at which a liquid boils at atmospheric pressure." The boiling points of hydrocarbons are related to the number of carbon atoms. The more carbon atoms, the higher the boiling point. The difference in boiling points makes possible the separation of different hydrocarbons (distillation and fractionation).

Boiling Point and Pressure

The boiling point of a compound is affected by the pressure at which the compound is held, e.g., propane boils at –43.8°F @ atmospheric pressure; however, if contained in a system at about 174 psi, the boiling point rises to +95°F.

Vapor Pressure

Any liquid is made up of many molecules, all of which are in motion. Some of these molecules have enough energy to break through the liquid's surface into the gaseous state. Consider an open container partially filled with liquid.

- The molecules that break free of the liquid can escape completely and the liquid will eventually evaporate.
- However, in a closed container, the increasing number of gaseous molecules creates an increase in pressure. It also increases the tendency for the molecules to re-enter the liquid.

Equilibrium is reached when the number of molecules leaving the liquid state equals the number re-entering the liquid. The pressure developed at this equilibrium is known as the *vapor pressure* of the liquid.

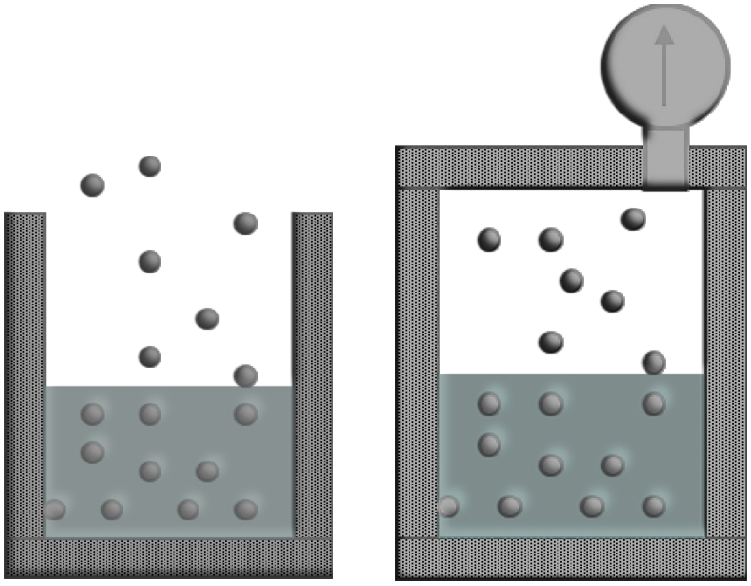


Figure 2: Open/Closed Container

Vapor Pressure and Temperature

The relationship between vapor pressure and temperature can be understood in terms of molecules in the liquid escaping the attractive forces of the liquid to enter the gas phase.

Boiling Point and Vapor Pressure

When a liquid is heated in an open vessel, the ***vapor pressure*** will increase until it reaches the atmospheric pressure surrounding the liquid. When this happens, it is accompanied by the physical phenomena of boiling, as vapor bubbles form in the liquid and rise to the surface.

The ***boiling point*** of a liquid is "the temperature at which its vapor pressure is equal to the surrounding pressure."

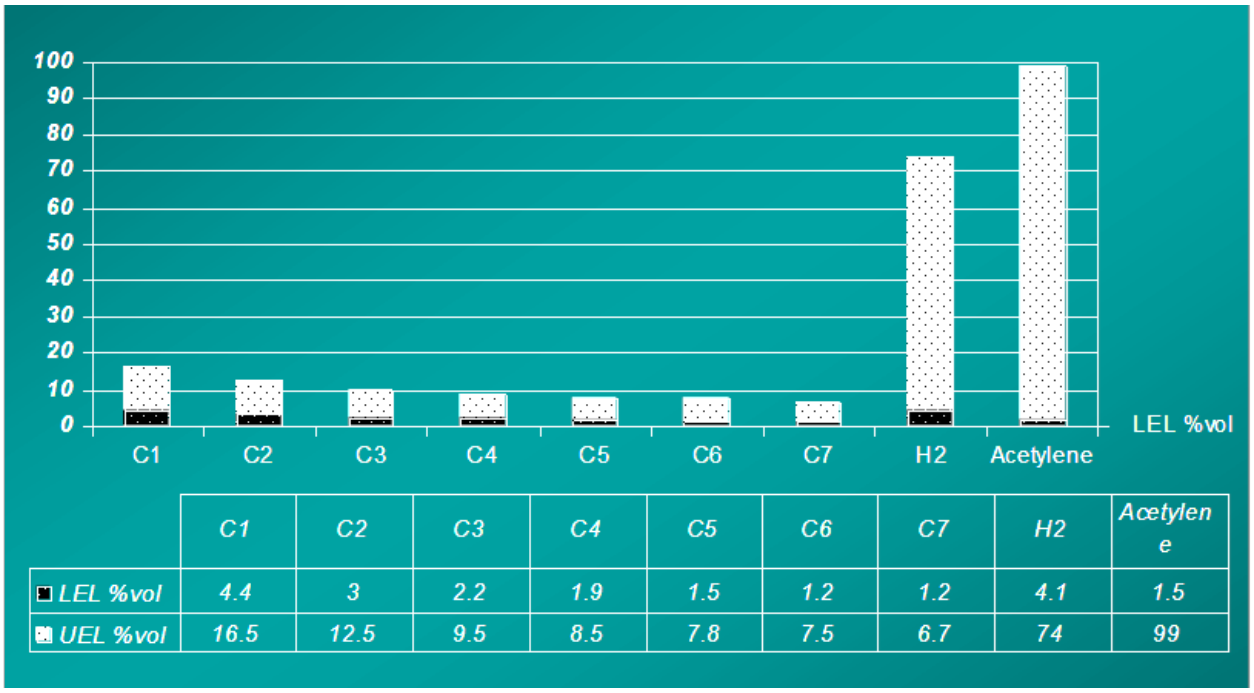
Specific Gravity

Specific gravity of a liquid is "a ratio of the weight of a volume of the liquid to the weight of an equal volume of water at a designated temperature."

Specific gravity of a gas is "a ratio of the weight of a volume of the gas to the weight of an equal volume of air at a designated temperature and pressure."

Table 1: Specific Gravity

Parameter	Methane C1 (CH_4)	Ethane C2 (C_2H_6)	Propane C3 (C_3H_8)	Iso-Butane iC4 (C_4H_{10})	N-Butane nC4 (C_4H_{10})
Molecular Weight	16	30	44	58	58
Liquid Specific Gravity	0.3	0.36	0.51	0.56	0.58
Vapor Specific Gravity	0.55	1.04	1.52	2.01	2.01



	C1	C2	C3	C4	C5	C6	C7	H2	Acetylene
LEL %vol	4.4	3	2.2	1.9	1.5	1.2	1.2	4.1	1.5
UEL %vol	16.5	12.5	9.5	8.5	7.8	7.5	6.7	74	99

Figure 3: Flammability Limits of Hydrocarbons

Table 2: Physical Properties of Hydrocarbons

Parameter	C1 Methane	C2 Ethane	C3 Propane
Chemical Formula	**CH_4**	C_2H_6	$C3H_8$
Molecular Weight	16	30	44
Boiling Point	**–259.6°F**	–128.2°F	–43.6°F

All the above hydrocarbons have a general formula of CnH2n+2 and belong to the classification of hydrocarbons called “paraffins.”

States of Matter

All matter exists in one of three states: solid, liquid, or a gas (refer to **Figure 4**).

Learning Objective 3

Name the three states of matter.

1. In a **solid**, molecules are close together and are slow-moving.
2. In a **liquid**, molecules are further apart and move faster than in a solid.
3. In a **gas**, molecules are far apart and move very rapidly.

The rate of molecule movement determines whether matter is a solid, a liquid, or a gas.

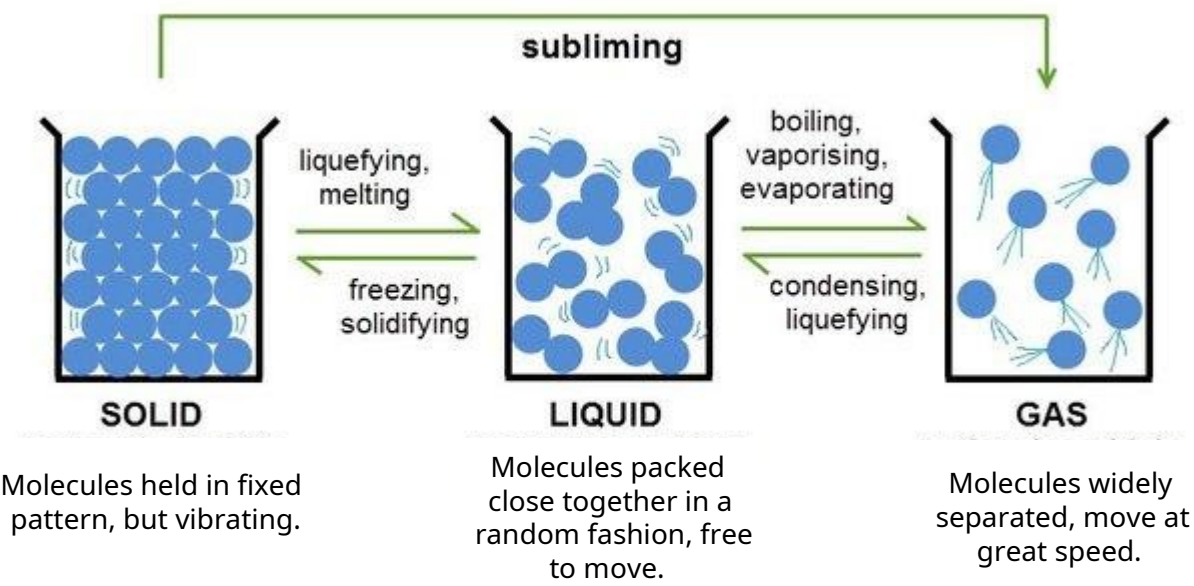


Figure 4: Three States of Matter

- **Solids** do not need a container as they have a definite shape and volume of their own.
- **Liquids** have a definite volume, but no definite shape. Liquids take the shape of whatever container they are put in.
- **Gases** have no definite shape or volume and expand to fill a container, no matter what its size or shape. A gas not enclosed in a container keeps on expanding until its molecules become so far apart, it is impossible to detect the gas.

Properties of Gases

Learning Objective 4

Describe the properties of gases.

Gas is a state of matter that has neither definite shape nor volume but tends to expand indefinitely.

The properties of a gas are Volume, Pressure, and Temperature.

The property changes of a gas are explained by the *Ideal Gas Laws*, which are briefly reviewed in this section. These are called the *Ideal Gas Laws* because no actual gas obeys it exactly, although all obey it over a wide range of conditions.)

Boyle's Law

Boyle's Law states that, for any *isothermal* (same temperature) process, **the volume of a fixed mass of gas is inversely proportional to its absolute pressure"** (refer to **Figure 5**).

$$P \ \alpha \ \frac{1}{V} \qquad \textbf{OR} \qquad PV = \text{a constant (F)}$$

Remember that all units should be 'absolute' and that:

Absolute pressure = Gauge pressure + Atmospheric pressure

Simply stated, *Boyle's Law* states that: **At constant temperature, if the volume of a gas is reduced by half, its pressure is doubled.**

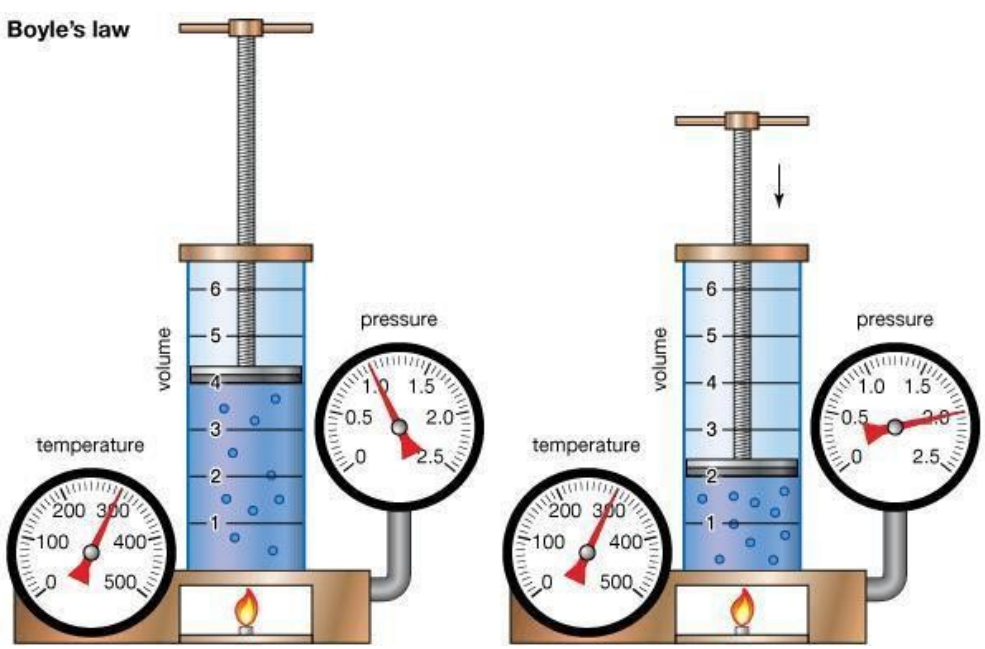


Figure 5: Boyle's Law

Flammability Comparison

A comparison to the properties of LNG to those of other liquid fuels, as shown in **Table 3**, also indicates that the lower flammability limit of LNG is generally higher than other fuels. That is, more LNG vapors would be needed (in a given area) to ignite as compared to LPG or gasoline.

Table 3: Comparison of LNG Properties to Other Liquid Fuels

PROPERTIES	LNG	LIQUEFIED PETROLEUM GAS (LPG)	GASOLINE	FUEL OIL
Toxic	No	No	Yes	Yes
Carcinogenic	No	No	Yes	Yes
Flammable Vapor	Yes	Yes	Yes	Yes
Forms Vapor Cloud	Yes	Yes	Yes	Yes
Asphyxiant	Yes, but in a vapor cloud	Yes, but in a vapor cloud	Yes	Yes
Extreme Cold Temperature	Yes	Yes, if refrigerated	No	No
Other Health Hazards	None	None	Eye irritant, narcosis (unconsciousness), nausea, others	Eye irritant, narcosis (unconsciousness), nausea, others
Flash Point (°F)	-306	-156	-50	140
Boiling Point (°F)	-256	-44	90	400
Flammability Range in Air (%)	5 – 15	2.1 – 9.5	1.3 – 6	N/A
Stored Pressure	Atmospheric	Pressurized (atmospheric if refrigerated)	Atmospheric	Atmospheric
Behavior if Spilled	Evaporates, forming visible "clouds." Portions of cloud could be flammable or explosive under certain conditions	Evaporates, forming visible "clouds." Portions of cloud could be flammable or explosive under certain conditions	Evaporates, forms flammable pool: Environmental cleanup required.	Evaporates, forms combustible pool: Environmental cleanup required.